Texas Tubular Products
P.O. Box 0388, 3681 FM 250, Lone Star, TX 75668

Print Date: 4/18/2024

Material Test Report

Results relate only to items tested. Test report not to be reproduced except in it's entirety.

Heat: B317535 **Product:** 8.625 0.322 X52 M PSL2 HFW **Mtr#:** 15

Lot: 57 **Specification:** API 5L 46th Edition (05/01/19) X52 M, PSL2
ASTM A53/A53M-22 Grade B/ASME SA-53/SA-53M-17 Grade B, ERW
HARDNESS MEETS NACE MRO175.
HFW MANUFACTURED IN THE U.S.A.

Customer Order Number: STL-306371

Sold To:
EDGEN MURRAY CORPORATION

BATON ROUGE LA 70884

Ship To:
EDGEN MURRAY CORPORATION

Chemical Analysis, %

Acceptance Criteria % Max CEPcm = .25

	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	V	B	Nb	Ti	Ca	N	CEIIW	CEPcm
Heat	0.06	0.64	0.013	0.0040	0.22	0.14	0.04	0.060	0.010	0.001	0.0001	0.028	0.018	0.0018	0.006	0.193	0.11
Product	0.06	0.62	0.011	0.0030	0.22	0.14	0.04	0.060	0.010	0.001	0.0001	0.020	0.010	0.0010	0.000	0.190	0.11
Product	0.05	0.62	0.010	0.0030	0.22	0.13	0.03	0.060	0.010	0.001	0.0001	0.020	0.010	0.0010	0.000	0.178	0.10
Product																	

Mechanical Properties

Test	Dir	Loc	Yield KSI	Tensile KSI	Elong % in 2"	Fracture Location	Y/T Ratio
1	T	B-180	65.9	74.9	37.4		0.88
2	T	W		78.2		OW	

Hydrotest, PSI (5 sec. hold) 3900
Flattening Passed
Specimen Size 1.5
Heat Treat Type Seam Normalization

Hardness Tests

Scale	Loc	Hard 1
HRBW	SUR	84.1

Charpy Impact Tests

Acceptance Criteria Min. Ft/Lb 13

Test	Dir	Loc	Size	Temp	Energy 1	Energy 2	Energy 3	Avg	% Shear 1	%Shear 2	%Shear 3	Avg
1	T	B-90	2/3	32 F	112	23	113	112	100	100	100	100
2	T	W	2/3	32 F	56	63	102	56	55	60	100	55

Remarks

MINIMUM HEAT TREATMENT OF WELD SEAM 1600 DEG F
TENSILE TEST - STRIP
MEETS ISO 10424 3.1.B / EN 10204 3.1 / LATEST EDITIONS
COIL MELTED AND ROLLED AT: SDI-Columbus, MS
MANUFACTURED BY TEXAS TUBULAR PRODUCTS LONE STAR, TX

Inspection Performed

ULTRASONIC WELDLINE - N10 Notch
VISUAL

By: Ryan Wauthier
Quality Manager

This is to certify that the product described herein was manufactured, sampled, tested, and /or inspected in accordance with the specifications/order and fulfills the requirements in such respects.

WE HEREBY VERIFY THAT THE ABOVE INFORMATION IS CORRECT AS CONTAINED IN THE RECORDS OF TEXAS TUBULAR PRODUCTS

Texas Tubular Products

P.O. Box 0388, 3681 FM 250, Lone Star, TX 75668

Print Date: 6/7/2024

Material Test Report

Results relate only to items tested. Test report not to be reproduced except in it's entirety.

Heat: B317535
Lot: 58
Product: 8.625 0.322 X52 M PSL2 HFW
Specification: API 5L 46th Edition (05/01/19) X52 M, PSL2
ASTM A53/A53M-22 Grade B/ASME SA-53/SA-53M-17 Grade B, ERW
HARDNESS MEETS NACE MRO175.
HFW MANUFACTURED IN THE U.S.A.
Mtr#: 14
Customer Order Number: STL-306371

Sold To:
EDGEN MURRAY CORPORATION

BATON ROUGE LA 70884

Ship To:
EDGEN MURRAY CORPORATION

Chemical Analysis, %

Acceptance Criteria % Max CEPcm = .25

	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	V	B	Nb	Ti	Ca	N	CEIIW	CEPcm
Heat	0.06	0.64	0.013	0.0040	0.22	0.14	0.04	0.060	0.010	0.001	0.0001	0.028	0.018	0.0018	0.006	0.193	0.11
Product	0.05	0.62	0.010	0.0030	0.22	0.13	0.03	0.060	0.010	0.001	0.0001	0.020	0.010	0.0010	0.000	0.178	0.10
Product	0.06	0.62	0.011	0.0030	0.22	0.14	0.04	0.060	0.010	0.001	0.0001	0.020	0.010	0.0010	0.000	0.190	0.11
Product																	

Mechanical Properties

Test	Dir	Loc	Yield KSI	Tensile KSI	Elong % in 2"	Fracture Location	Y/T Ratio
1	T	B-180	65.7	75.5	38.3		0.87
2	T	W		81.5		OW	

Hydrotest, PSI (5 sec. hold) 3900
Flattening Passed
Specimen Size 1.5
Heat Treat Type Seam Normalization

Hardness Tests

Scale	Loc	Hard 1
HRBW	SUR	82.9

Charpy Impact Tests

Acceptance Criteria Min. Ft/Lb 13

Test	Dir	Loc	Size	Temp	Energy 1	Energy 2	Energy 3	Avg	% Shear 1	%Shear 2	%Shear 3	Avg
1	T	B-90	2/3	32 F	118	124	123	118	100	100	100	100
2	T	W	2/3	32 F	66	119	88	66	100	100	100	100

Remarks

MINIMUM HEAT TREATMENT OF WELD SEAM 1600 DEG F
TENSILE TEST - STRIP
MEETS ISO 10424 3.1.B / EN 10204 3.1 / LATEST EDITIONS
COIL MELTED AND ROLLED AT: SDI-Columbus, MS
MANUFACTURED BY TEXAS TUBULAR PRODUCTS LONE STAR, TX

Inspection Performed

ULTRASONIC WELDLINE - N10 Notch
VISUAL

By: Ryan Wauters
Quality Manager

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